

Wanderlust Categorized: Data-Driven Customer Segmentation for TravelTide



*Blending business rules with machine learning to
drive smarter rewards and personalization.*

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Project Overview

Goal

Segment TravelTide customers to design a targeted rewards program and enhance marketing strategies.

Why it matters

Personalization boosts customer satisfaction, loyalty, and conversion rates.

Approaches

Rule-Based Segmentation
(based on known behavior patterns)

ML-Based Segmentation
(unsupervised clustering to uncover hidden groups)

Project Steps

Exploratory Data Analysis

Cleaned data and filtered 50,000 sessions for exploration to understand the users and explore patterns.

Rule-Based Groups

Created rule based segments based on insights derived from exploratory data analysis.

Machine Learning Algorithms

Applied PCA for dimensionality reduction and used K-Means and DBSCAN to create clusters in the data.

Cohort Selection

Selected a cohort to use for ML clustering algorithms, by querying an user-based table with SQL.

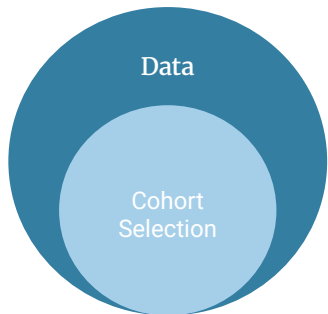
Sessions older than 2023-01-05
Users with more than 7 sessions

Feature Engineering and Preprocessing

Created new features and prepared the data for ML clustering algorithms.

User Groups and Perks

Created personas for each group with perks.



Rule Based Segmentation

1	Single Adventurer	• 18+ • Not married and no children • Open to explore and meet new people	5	Business Traveler	• More than 4 trips • Less than 2 bags and 2 rooms • Prioritizes efficiency, flexibility, and convenience
2	Couples	• 30 - 59 • Married and no children • Often travel for romantic getaways or city breaks	6	Senior Couple	• 60+ • Married • Values comfort, safety, and relaxing experiences
3	Families	• 30+ • Married with children • Need convenience, affordability, and child-friendliness.	7	Young Adventurer	• Younger than 18 • No children and not married • Seeking experiences, often budget-focused
4	Single Parent	• 19+ • Not married with children • Budget-aware, time-stressed, needs practical perks	8	Senior Single	• 60+ • Not married • May be solo-traveling for leisure or to visit family





Lovebirds on Leave Couples

“Just the two of us,
and a hot tub please!”

✓ Offer a romantic package (wine, flowers, spa) or weekend deals



Golden Getaway Senior Couple

“Retired from work,
not from wonder!”

✓ Offer a room upgrade to a quiet floor, or with a scenic view

The Nesters Families

“Family is all you need for
the best holiday
memories!”

✓ Offer free stay for children or a child friendly entertainment package (partner deals)



Power Parent Single Parent

“One parent, many
hats, no time to waste!”

✓ Offer flexible booking policies or hotel babysitting





Freewheel Solo

Single Traveler (>18)

“One backpack. No plan.
All in.”

✔ Offer public transport pass or discounted solo tours



Jet-Set Pro

Business Traveler

“In by 9, out by 5 — and onto
the next city.”

✔ Offer flight upgrades or access to airport lounge

Trailblazer in Training

Young Traveler (<18)

“Too young to rent a car, but not
too young to explore the world.”

✔ Offer discounted adventure or social group tours



Silver Explorer

Single Senior (>60)

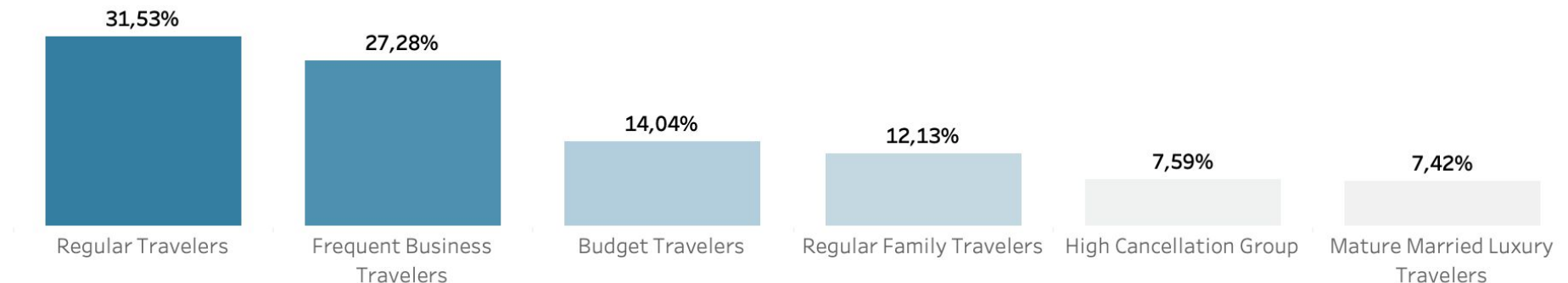
“One ticket, one heart, endless
stories.”

✔ Offer curated tours or emergency contact services



K-Means Segmentation

1	Regular Travelers	• Average age is 42 • Not married without children • Regular, consistent travel patterns	4	Regular Family Travelers	• Average age is 47 • Married with children • Moderate travel behavior • Highest average of checked bags
2	Frequent Business Travelers	• Average age is 44 • Highest trips, flights, and spending • Frequent flyers, value efficiency	5	High Cancellation Group	• Average age is 43 • Not married without children • Extremely high cancellation rate • Low follow-through on bookings
3	Budget Travelers	• Average age is 32 • Not married without children • Fewest trips and lowest spending	6	Mature Married Luxury Travelers	• Average age is 57 • Married without children • Fewer trips, longer stays, high average spending,





The Steady Explorers

Regular Traveler

“I love planning spontaneous getaways — not too fancy, just new experiences and a change of scenery.”

✓ Offer targeted destination packages



The Spontaneous Savers

Budget Traveler

“I travel when the deals are good and the vibe feels right — I’m always up for an adventure, as long as it’s cheap!”

✓ Offer exclusive discounts or Last-minute deal alerts



The Executive Jetsetters

Business Travelers

“Efficiency, comfort, and staying connected — travel is part of the job, but I expect it to run smoothly.”

✓ Offer airport lounge access or Express check-in/security lane access



The Family Explorers

Regular Family
Travelers

“Traveling with the kids isn’t just a holiday — it’s how we make memories that last a lifetime.”

✅ Offer free checked bags
(they have highest average
checked bags)



The Golden Indulgers

Mature Married Luxury
Travelers

“We prefer fewer trips, but each one is special — comfort, fine dining, and time to unwind together.”

✅ Offer a free hotel night on
longer stays (they already stay
longest)



The Uncommitted Clickers

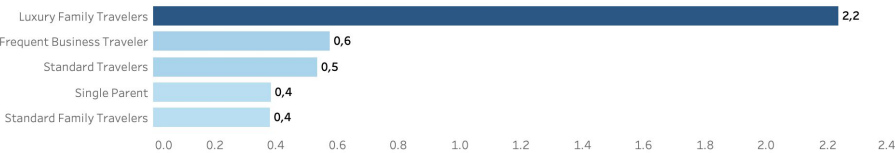
High Cancellation Group

“Life changes fast — I book when I can, but I need the flexibility to change my plans without stress.”

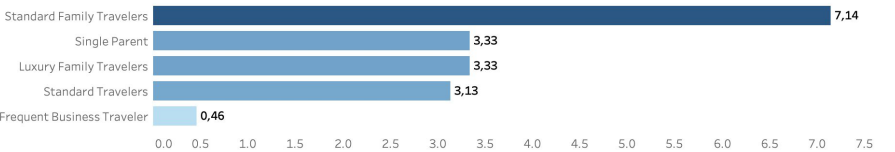
✅ Offer free cancellation

DBSCAN Segmentation

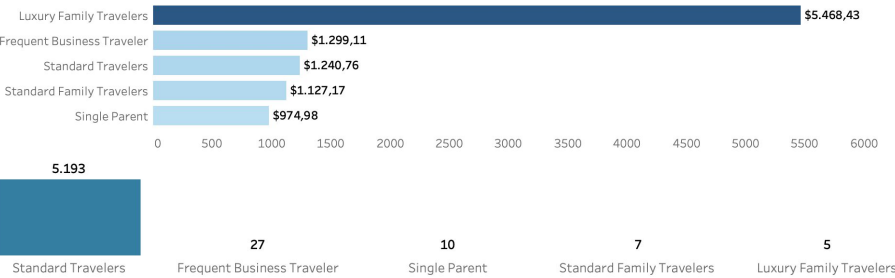
Average Checked Bags



Average Cancel Rate



Average Price Paid



1	Standard Travelers	- Average age is 43 - Not married without children - Average trips, flights, spending, and stay
2	Frequent Business Travelers	- Average age is 45 - Highest trips, flights, and spending - Frequent flyers, low cancellation
3	Single Parents	- Average age is 40 - Not married with children - Moderate trips and flights with low spending
4	Standard Family Travelers	- Average age is 45 - Married with children - Family travelers with high tendency to cancel
5	Luxury Family Travelers	- Average age is 37 - Married with children - Highest spending with the most average checked bags

Conclusion and Recommendation

Conclusion

- 📌 Rule-based groups helped frame customer types
- 📌 K-means provides superior business value
- 📌 DBSCAN's strong mathematical clustering doesn't translate to actionable insights

Recommendations

- ✅ Launch Personalized Perks Program based on the 6 K-means segments
- ✅ Prioritize High-Value Segments and develop Cancellation strategies
- ✅ Create VIP Treatment for the 5 luxury outliers identified by DBSCAN
- ✅ Conduct Retention Interventions for at-risk customers in smaller DBSCAN clusters
- ✅ Re-evaluate segment and perk performance quarterly

Thank you for your attention!

